



FINAL MARK

GIRRAWEE HIGH SCHOOL
YEAR 11
MATHEMATICS (ADVANCED)
2019 Yearly Examination

Name: _____

QUESTION	MARK	MA11-1	MA11-2	MA11-3	MA11-4	MA11-5	MA11-6	MA11-9
1-5	/5	✓	✓					✓
6	/18	✓						✓
7	/26	✓				✓		✓
8	/19	✓	✓				✓	✓
9	/20	✓	✓				✓	✓
10	/12	✓	✓					✓
11	/13	✓	✓	✓				✓
12	a,d/8	✓	✓	✓	✓			✓
12	b,c/7	✓	✓	✓				✓
TOTAL								
	/128	/128	/84	/28	/8	/26	/39	/128

Year 11 Mathematics Advanced outcomes

A student:

- MA11-1** uses algebraic and graphical techniques to solve, and where appropriate, compare alternative solutions to problems
- MA11-2** uses the concepts of functions and relations to model, analyse and solve practical problems
- MA11-3** uses the concepts and techniques of trigonometry in the solution of equations and problems involving geometric shapes
- MA11-4** uses the concepts and techniques of periodic functions in the solutions of trigonometric equations or proof of trigonometric identities
- MA11-5** interprets the meaning of the derivative, determines the derivative of functions and applies these to solve simple practical problems
- MA11-6** manipulates and solves expressions using the logarithmic and index laws, and uses logarithms and exponential functions to solve practical problems
- MA11-7** uses concepts and techniques from probability to present and interpret data and solve problems in a variety of contexts, including the use of probability distributions
- MA11-8** uses appropriate technology to investigate, organise, model and interpret information in a range of contexts
- MA11-9** provides reasoning to support conclusions which are appropriate to the context



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MATHEMATICS (ADVANCED)
YEAR 11
Yearly Examination 2019

Year 11 Mathematics

Time allowed: 120 minutes (plus10 minutes reading time)

Instructions:

- Attempt ALL questions.
- For Questions 1 to 5, colour in the bubble next to the letter corresponding to the correct response on your paper.
- For Questions 6 to 12:
 - All necessary working should be shown in the space provided.
 - Marks will be deducted for careless or badly arranged work.
 - Board-approved calculators may be used.
 - Reference sheet is provided.
 - Diagrams are NOT to scale.

Section 1 (5 marks)

Use the multiple-choice answer sheet for questions 1-5.

Multiple Choice (5 marks).

1. The simplest form of the expression $\frac{1}{\sqrt{3}-\sqrt{2}} - \sqrt{3}$ is:

- A) $\sqrt{2}$ B) $\sqrt{3}$ C) $\sqrt{3} + \sqrt{2}$ D) $\sqrt{3} - \sqrt{2}$

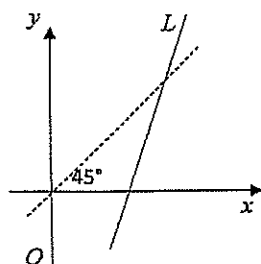
2. The simplest form of the algebraic fraction $\frac{n}{n-1} + \frac{n}{1+n}$ is :

- A) 1 B) -1 C) 0 D) $\frac{2n^2}{n^2-1}$

3. The degree of the polynomial $(x^5 + 1)^2$, is

- A) 2 B) 5 C) 7 D) 10

4) Given the line L in the diagram below, the gradient of the line is m . Which one of the following is correct about m .



- A) $m = 1$ B) $m = 0$ C) $m < 1$ D) $m > 1$

5. What is the derivative of e^{x^2} .

- A) $x^2 e^{x^2-1}$ B) $2xe^{2x}$ C) $2xe^{x^2}$ D) $2e^{x^2}$

Answers to Multiple Choice Questions

Select the alternative A, B, C or D that best answers the question.

1.	A	☐	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☐
3.	A	☐	B	☐	C	☐	D	☐
4.	A	☐	B	☐	C	☐	D	☐
5.	A	☐	B	☐	C	☐	D	☐

Section 11

- Answer questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 6 (18 marks)

a) Find the value of $4\pi\sqrt{\frac{a}{g}}$ if $a=4.1$ and $g=9.8$. Give your answer to 2 significant figures.
 1

b) Write down the value of $|-9| - |-10|$
1

c) Factorise fully:

i) $x^2 + 5x - 6$
 1

ii) $2x^2 - 7x - 15$
 2

d) Simplify:

i) $\frac{2x^2 - xy}{4x^2 - y^2}$ 2

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ii) $\frac{2}{3} - \frac{x-1}{4}$ 2

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iii) $\frac{1}{x^2-4} - \frac{2}{x-2} - \frac{3}{x+2}$ 3

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e) Express $\frac{21}{\sqrt{63}} - \frac{3}{\sqrt{7}+2}$ as an integer. (show necessary working) 3

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f) If $a = \left(\frac{1}{5}\right)^{\frac{1}{3}}$ and $b = \left(\frac{1}{5}\right)^{\frac{1}{2}}$, evaluate the expression $\frac{a^3 + b^2}{(ab)^6}$. 3

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Question 7 (26 marks)

(a) Solve :

i) $3(2x+1) - 2(3-x) = 53$

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ii) $3x^2 - 4x - 4 = 0$

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iii) $16^x = 64$

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iv) $|4x-3|=7$

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$$v) \frac{x-5}{3} - \frac{x+1}{4} = 5$$

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b) Differentiate :

$$i) y = \frac{7}{2x}$$

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$$ii) y = \frac{5-x^2+5x^4}{x^2}$$

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$$iii) y = 5(4-x)^9$$

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iv) $y = x^2(3x+2)^3$

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v) $y = \frac{x^2-4}{x-1}$

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Question 8 (19 marks).

a) Differentiate $y = \frac{2x}{e^x+1}$

3

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b) i) Differentiate $y = e^x - x$

1

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ii) For what value of x is the curve stationary? 2

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c) i) Find the gradient of the tangent to $y = e^{-x}$ at the point $P(-1, e)$. 2

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ii) Hence write down the gradient of the normal at this point. 1

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iii) Hence determine the equation of this normal. 2

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iv) Find the x and y intercepts of the normal. 2

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v) Find the area of the triangle whose vertices lie at the intercepts and the origin. 3

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d) Find the equation of the normal to the curve $y = \sqrt{x+2}$ at the point (7,3). 3

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Question 9 (20 marks).

a) Evaluate
i) $\log_{10} 875$ correct to 2 decimal places. 1

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ii) Evaluate $\log_e 80$ to 3 significant figures. 1

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iii) $\log_5 2$ correct to 2 decimal places 2

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b) Evaluate:

i) $\log_2 16$ 2

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ii) $\log_5 25 + 1$ 2

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iii) $\log_3 \frac{1}{\sqrt[3]{3}}$
2

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c) Given $\log_5 3 = 0.68$ and $\log_5 4 = 0.86$, find

i) $\log_5 12$
1

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ii) $\log_5 0.75$
1

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iii) $\log_5 9$
2

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iv) $\log_5 20$
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d) The amount of money A in a bank account after n years grows with compound interest according to the formula $A = 850(1.025)^n$.

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i) Find the initial amount in the bank. 1

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ii) Find the amount after 7 years. 1

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iii) Find how many years it will take for the amount in the bank to be \$1000. 2

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Question 10 (12 marks)

a) Show that the lines $5x - y - 10 = 0$, $x + y - 8 = 0$ and $2x - 3y + 9 = 0$ are concurrent? 3

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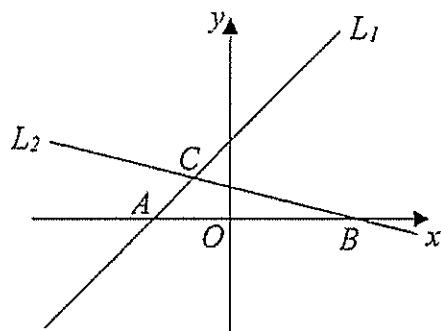
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b) In the diagram below, the line $L_1: x - y + 2 = 0$ cuts the x axis at A and the line $L_2: x + 4y - 3 = 0$ cuts the x axis at B .



i) Show that the co-ordinates of A and B are $(-2,0)$ and $(3,0)$ respectively.

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ii) If C is the point of intersection of the lines L_1 and L_2 , use simultaneous equations to show that the co-ordinates of C are $(-1,1)$.

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iii) Find the length of AC .

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iv) Calculate the gradient of the line AC .

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v) Show that the line L_1 makes an angle of 45° with the positive direction of the x -axis.

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vi) Find the area of $\triangle ABC$.

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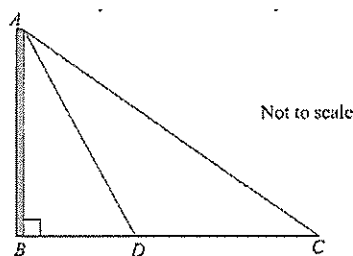
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Question 11 (13 marks).

- a) A vertical tower AB with points B, C and D in a straight line on the ground is shown below. The distance from CD is 100 metres. The angle of elevation to the top of the tower from point C is 35° and from point D is 60° .



- i) Show that $AD = \frac{100 \sin 35^\circ}{\sin 25^\circ}$ 3

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- ii) Calculate the height of the tower. Answer to the nearest metre. 2

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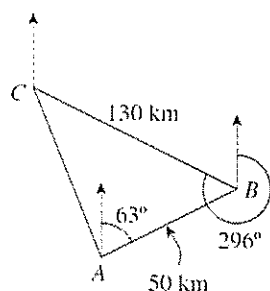
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b) A ship sails 50 km from port A to port B on a bearing of $063^\circ T$, then sails 130 km from port B to port C on a bearing of $296^\circ T$.



i) Show that $\angle ABC = 53^\circ$

2

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ii) Find, correct to the nearest km , the distance of port A from port C .

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iii) Use the cosine rule to find $\angle ACB$, and hence find the bearing of port A from port C , correct to the nearest degree.

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Question 12.(15 marks).

a) Prove that $\tan \theta - \tan \theta \sin^2 \theta = \sin \theta \cos \theta$.

2

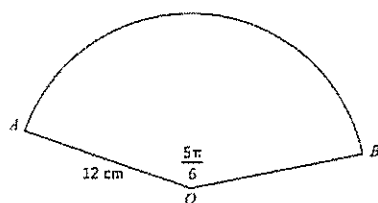
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b) In the diagram below,



AB is an arc of a circle with centre O . The radius $OA = 12cm$ and $\angle AOB = \frac{5\pi}{6}$ radians.

i) Find the length of the arc AB .

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ii) Find the area of the sector AOB .

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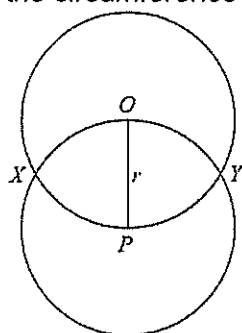
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- c) Two equal circles with centres O and P intersect at X and Y as shown below. The centres of each circle lie on the circumference of the other circle.



Find the exact area of the region $XOYP$. (Answer in terms of r)

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- d) Solve these trigonometric equations in radians, for $0 \leq x \leq 2\pi$.

i) $\sin^2 x = \frac{3}{4}$

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ii) $2 \cos^2 x - \cos x - 1 = 0$

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End of Examination



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Section 1 (5 marks)

Use the multiple-choice answer sheet for questions 1-5.

Multiple Choice (5 marks).

1. The simplest form of the expression $\frac{1}{\sqrt{3}-\sqrt{2}} - \sqrt{3}$ is: $\frac{\sqrt{3}+\sqrt{2}}{(\sqrt{3}-\sqrt{2})(\sqrt{3}+\sqrt{2})} - \sqrt{3} = \frac{\sqrt{3}+\sqrt{2}}{3-2} - \sqrt{3} = \sqrt{3}+\sqrt{2}-\sqrt{3} = \sqrt{2}$

A) $\sqrt{2}$

B) $\sqrt{3}$

C) $\sqrt{3}+\sqrt{2}$

D) $\sqrt{3}-\sqrt{2}$

2. The simplest form of the algebraic fraction $\frac{n}{n-1} + \frac{n}{1+n}$ is: $\frac{n+n^2+n^2-n}{n^2-1} = \frac{2n^2}{n^2-1}$

A) 1

B) -1

C) 0

D) $\frac{2n^2}{n^2-1}$

3. The degree of the polynomial $(x^5+1)^2$, is

A) 2

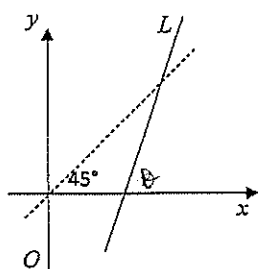
B) 5

C) 7

D) 10

4) Given the line L in the diagram below, the gradient of the line is m .

Which one of the following is correct about m .



$\theta > 45^\circ$ (ext. $\angle >$ interior oppo.)
 $\therefore m = \tan \theta > \tan 45^\circ$
 $m > 1$

A) $m = 1$

B) $m = 0$

C) $m < 1$

D) $m > 1$

5. What is the derivative of e^{x^2} .

$$\frac{d}{dx} e^{x^2} = e^{x^2} \times 2x$$

A) $x^2 e^{x^2-1}$

B) $2xe^{2x}$

C) $2xe^{x^2}$

D) $2e^{x^2}$

Answers to Multiple Choice Questions

Select the alternative A, B, C or D that best answers the question.

1.	A	☒	B	☐	C	☐	D	☐
2.	A	☐	B	☐	C	☐	D	☒
3.	A	☐	B	☐	C	☐	D	☒
4.	A	☐	B	☐	C	☐	D	☒
5.	A	☐	B	☐	C	☒	D	☐

Section 11

- Answer questions in the spaces provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Question 6 (18 marks)

a) Find the value of $4\pi\sqrt{\frac{a}{g}}$ if $a=4.1$ and $g=9.8$. Give your answer to 2 significant figures.

$$\dots\dots\dots 4\pi\sqrt{\frac{4.1}{9.8}} = 8.11 \dots\dots\dots (1)$$

b) Write down the value of $|-9| - |-10|$

$$\dots\dots\dots = 9 - 10 = -1 \dots\dots\dots (1)$$

c) Factorise fully:

i) $x^2 + 5x - 6$

$$\dots\dots\dots (x+6)(x-1) \dots\dots\dots (1)$$

$$\begin{aligned} \text{ii) } 2x^2 - 7x - 15 &= (2x^2 - 10x) + (3x - 15) \\ &= 2x(x-5) + 3(x-5) \\ &= (x-5)(2x+3) \end{aligned} \dots\dots\dots (2)$$

d) Simplify:

i) $\frac{2x^2 - xy}{4x^2 - y^2}$

$$\dots\dots\dots = \frac{x(2x-y)}{(2x-y)(2x+y)} = \frac{x}{2x+y} \dots\dots\dots (2)$$

ii) $\frac{2}{3} - \frac{x-1}{4}$

$$\begin{aligned} \dots\dots\dots &= \frac{8 - 3(x-1)}{12} = \frac{8 - 3x + 3}{12} \\ &= \frac{11 - 3x}{12} \end{aligned} \dots\dots\dots (2)$$

$$\text{iii) } \frac{1}{x^2-4} - \frac{2}{x-2} - \frac{3}{x+2} = \frac{1}{(x-2)(x+2)} - \frac{2}{x-2} - \frac{3}{x+2}$$

$$= \frac{1 - 2(x+2) - 3(x-2)}{(x-2)(x+2)} = \frac{1 - 2x - 4 - 3x + 6}{(x-2)(x+2)}$$

$$= \frac{-5x + 3}{(x-2)(x+2)}$$

e) Express $\frac{21}{\sqrt{63}} - \frac{3}{\sqrt{7}+2}$ as an integer.

3

$$\begin{aligned} &= \frac{21 \times \sqrt{63}}{63} - \frac{3(\sqrt{7}-2)}{(\sqrt{7}+2)(\sqrt{7}-2)} = \frac{\sqrt{63}}{3} - \frac{3(\sqrt{7}-2)}{3} \\ &= \frac{(\sqrt{63} - 3\sqrt{7} + 6)}{3} \\ &= \frac{3\sqrt{7} - 3\sqrt{7} + 6}{3} = \frac{6}{3} = 2 \end{aligned}$$

f) If $a = \left(\frac{1}{5}\right)^{\frac{1}{3}}$ and $b = \left(\frac{1}{5}\right)^{\frac{1}{2}}$, evaluate the expression $\frac{a^3 + b^2}{(ab)^6}$.

3

$$\begin{aligned} &= \frac{\frac{1}{5} + \frac{1}{5}}{\left(\left(\frac{1}{5}\right)^{\frac{5}{6}}\right)^6} = \frac{\frac{2}{5}}{\left(\frac{1}{5}\right)^5} = \frac{2}{5} \times \frac{5^5}{1} \\ &= 2 \times 5^4 = 1250 \end{aligned}$$

Question 7 (26 marks)

(a) Solve :

i) $3(2x+1) - 2(3-x) = 53$

2

$$\begin{aligned} 6x + 3 - 6 + 2x &= 53 \\ 8x - 3 &= 53 \\ 8x &= 56 \\ x &= 7 \end{aligned}$$

ii) $3x^2 - 4x - 4 = 0$

2

$$\begin{aligned} (3x+2)(x-2) &= 0 \\ \therefore x &= -\frac{2}{3} \text{ or } 2 \end{aligned}$$

iii) $16^x = 64$

2

$$\begin{aligned} 16^x &= 64 \\ 4^{2x} &= 4^3 \quad \therefore 2x = 3 \quad \therefore x = \frac{3}{2} \\ \text{OR} \quad 2^{4x} &= 2^6 \quad \therefore 4x = 6 \quad \therefore x = \frac{3}{2} \end{aligned}$$

iv) $|4x-3|=7$

3

$$\begin{aligned} 4x-3 &= 7 \quad \text{or} \quad -(4x-3) = 7 \\ 4x &= 10 \quad \text{or} \quad -4x+3 = 7 \\ x &= \frac{5}{2} \quad \text{or} \quad -4x = 4 \\ & \quad \quad \quad x = -1 \end{aligned}$$

$$v) \frac{x-5}{3} - \frac{x+1}{4} = 5$$

3

$$(x12) \dots\dots\dots 4(x-5) - 3(x+1) = 60$$

$$\dots\dots\dots 4x - 20 - 3x - 3 = 60$$

$$\dots\dots\dots x - 23 = 60$$

$$\dots\dots\dots x = 83$$

b) Differentiate :

$$i) y = \frac{7}{2x} = \frac{7}{2} x^{-1}$$

$$\frac{dy}{dx} = -\frac{7}{2} x^{-2} = -\frac{7}{2x^2}$$

2

$$ii) y = \frac{5-x^2+5x^4}{x^2} = \frac{5}{x^2} - 1 + 5x^2 = 5x^{-2} - 1 + 5x^2$$

$$\frac{dy}{dx} = -10x^{-3} + 10x = -\frac{10}{x^3} + 10x$$

3

$$iii) y = 5(4-x)^9$$

$$\frac{dy}{dx} = 5 \cdot x \cdot 9 \cdot (4-x)^8 \cdot (-1)$$

$$= -45(4-x)^8$$

3

$$iv) y = x^2(3x+2)^3$$

$$u = x^2 \quad | \quad v = (3x+2)^3$$

$$u' = 2x \quad | \quad v' = 3(3x+2)^2 \cdot 3$$

$$= 9(3x+2)^2$$

$$\frac{dy}{dx} = v \cdot u' + u \cdot v' = (3x+2)^3 \cdot (2x) + x^2 \cdot 9(3x+2)^2$$

$$= x(3x+2)^2 [6x+4+9x]$$

$$= x(3x+2)^2 [15x+4]$$

3

$$v) y = \frac{x^2 - 4}{x - 1}$$

$$u = x^2 - 4 \quad | \quad v = x - 1$$

$$u' = 2x \quad | \quad v' = 1$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{(x-1) \cdot 2x - (x^2 - 4) \cdot 1}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2 + 4}{(x-1)^2} = \frac{x^2 - 2x + 4}{(x-1)^2}$$

Question 8 (19 marks)

a) Differentiate $y = \frac{2x}{e^x + 1}$

3

$$u = 2x \quad | \quad v = e^x + 1$$

$$u' = 2 \quad | \quad v' = e^x$$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2} = \frac{(e^x + 1) \cdot 2 - 2x(e^x)}{(e^x + 1)^2}$$

$$= \frac{2e^x + 2 - 2xe^x}{(e^x + 1)^2}$$

b) i) Differentiate $y = e^x - x$

1

$$\frac{dy}{dx} = e^x - 1$$

ii) For what value of x is the curve stationary?

2

$$\text{stationary} \Rightarrow \frac{dy}{dx} = 0 \quad \begin{matrix} e^x - 1 = 0 \\ e^x = 1 \quad \therefore x = 0 \end{matrix}$$

b) Find the equation of the normal to the curve $y = \sqrt{x+2}$ at the point (7,3).

3

$$y = (x+2)^{\frac{1}{2}} \quad \therefore \frac{dy}{dx} = \frac{1}{2}(x+2)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x+2}}$$

$$\text{at } (7,3), \frac{dy}{dx} = \frac{1}{2\sqrt{9}} = \frac{1}{6} \quad \therefore m_{\text{normal}} = -6$$

$$\text{eqn: } y - 3 = -6(x - 7) \quad \therefore 6y - 18 = -6x + 42$$

$$y - 3 = -6x + 42$$

$$6x + y - 45 = 0$$

c) i) Find the gradient of the tangent to $y = e^{-x}$ at the point $P(-1, e)$.

2

$$\frac{dy}{dx} = -e^{-x}, \text{ at } (-1, e) \frac{dy}{dx} = -e^1 = -e$$

ii) Thus write down the gradient of the normal at this point.

1

$$\therefore \text{gradient of normal} = \frac{1}{e}$$

iii) Hence determine the equation of this normal.

2

$$y - e = \frac{1}{e}(x + 1)$$

$$ey - e^2 = x + 1$$

$$\therefore x - ey + e^2 + 1 = 0$$

iv) Find the x and y intercepts of the normal.

$$x - ey + e^2 + 1 = 0$$

2

$$x\text{-intercept} \Rightarrow y = 0 \therefore x = -(1 + e^2)$$

$$y\text{-intercept} \Rightarrow x = 0 \therefore y = \frac{e^2 + 1}{e}$$

v) Find the area of the triangle whose vertices lie at the intercepts and the origin.

3

$$\text{Area} = \frac{1}{2}bh = \frac{1}{2}(1 + e^2)\left(\frac{e^2 + 1}{e}\right)$$

$$= \frac{1}{2}(e^4 + 2e^2 + 1)$$

$$= \frac{1}{2}(e^3 + 2e + e^{-1})u^2$$

Question 9 (20 marks).

a) Evaluate

i) $\log_{10} 875$ correct to 2 decimal places.

1

..... $= 2.94$

ii) Evaluate $\log_e 80$ to 3 significant figures.

1

..... $= 4.38$

iii) $\log_5 2$ correct to 2 decimal places

2

..... $= \frac{\log_{10} 2}{\log_{10} 5} = 0.43$

b) Evaluate:

i) $\log_2 16$

2

..... $= \log_2 2^4 = 4 \log_2 2 = 4$

.....

ii) $\log_5 25 + 1$

2

..... $= \log_5 5^2 + 1 = 2 + 1 = 3$

.....

iii) $\log_3 \frac{1}{\sqrt[3]{3}}$

2

..... $= \log_3 3^{-\frac{1}{3}} = -\frac{1}{3} \log_3 3 = -\frac{1}{3}$

.....

.....

.....

c) Given $\log_5 3 = 0.68$ and $\log_5 4 = 0.86$, find

i) $\log_5 12$

1

..... $= \log_5 (3 \times 4) = \log_5 3 + \log_5 4$

..... $= 0.68 + 0.86 = 1.54$

ii) $\log_5 0.75$

1

$$\begin{aligned} &= \log_5 \frac{3}{4} = \log_5 3 - \log_5 4 \\ &= 0.68 - 0.86 \\ &= -0.18 \end{aligned}$$

iii) $\log_5 9$

2

$$\begin{aligned} &= \log_5 3^2 = 2 \log_5 3 \\ &= 2 \times 0.68 \\ &= 1.36 \end{aligned}$$

iv) $\log_5 20$

2

$$\begin{aligned} &= \log_5 (5 \times 4) = \log_5 5 + \log_5 4 \\ &= 1 + 0.86 = 1.86 \end{aligned}$$

d) The amount of money A in a bank account after n years grows with compound interest according to the formula $A = 850(1.025)^n$.

i) Find the initial amount in the bank. $A = 850(1.025)^n$

1

$$n=0 \Rightarrow A = \$850$$

ii) Find the amount after 7 years. $n=7$

1

$$A = 850(1.025)^7 = \$1010.35$$

iii) Find how many years it will take for the amount in the bank to be \$1000.

2

$$\begin{aligned} 1000 &= 850(1.025)^n \\ \frac{1000}{850} &= 1.025^n = \frac{20}{17} \\ n \log(1.025) &= \log\left(\frac{20}{17}\right) \end{aligned}$$

$$\therefore n = \frac{\log \frac{20}{17}}{\log 1.025} \approx 6.6$$

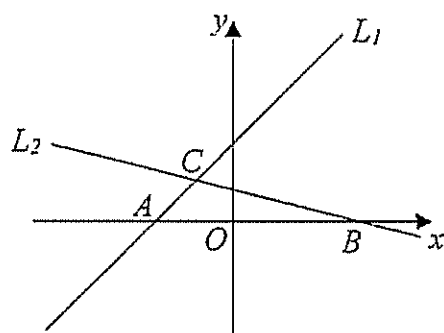
Question 10 (12 marks)

a) Are the lines $5x - y - 10 = 0$, $x + y - 8 = 0$ and $2x - 3y + 9 = 0$ concurrent?

3

$5x - y - 10 = 0 \quad \text{--- ①}$ $x + y - 8 = 0 \quad \text{--- ②}$ $\text{①} + \text{②} \quad 6x - 18 = 0$ $x = 3$ $\text{sub } x = 3 \text{ in ①} \quad 15 - y - 10 = 0 \quad \therefore y = 5$ $\text{These lines intersect at } (3, 5).$	$\left \begin{array}{l} \text{sub in } (3, 5) \text{ in} \\ \text{the line } 2x - 3y + 9 = 0 \\ \text{LHS: } 6 - 15 + 9 = 0 = \text{RHS} \\ \therefore \text{Three lines are} \\ \text{concurrent.} \end{array} \right.$
--	---

) In the diagram below, the line $L_1: x - y + 2 = 0$ cuts the x -axis at A and the line $L_2: x + 4y - 3 = 0$ cuts the x -axis at B .



i) Show that the co-ordinates of A and B are $(-2, 0)$ and $(3, 0)$ respectively.

2

co-ordinates of $A \Rightarrow y = 0 \quad x - y + 2 = 0 \quad \therefore x = -2 \quad \therefore A(-2, 0)$

co-ordinates of $B \Rightarrow y = 0 \quad \text{in } x + 4y - 3 = 0 \quad \therefore x = 3 \quad \therefore B(3, 0)$

ii) If C is the point of intersection of the lines L_1 and L_2 , use simultaneous equations to show that the co-ordinates of C are $(-1, 1)$.

2

$x - y + 2 = 0 \quad \text{--- ①}$ $x + 4y - 3 = 0 \quad \text{--- ②}$ $\text{①} - \text{②} \quad -5y + 5 = 0$ $\therefore y = 1$	$\left \begin{array}{l} \text{sub in } y = 1 \text{ in ①} \\ x - 1 + 2 = 0 \\ \therefore x = -1 \\ \therefore C(-1, 1) \end{array} \right.$
---	--

iii) Find the length of AC .

1

$$A(-2, 0), C(-1, 1)$$
$$AC = \sqrt{(-2+1)^2 + (0-1)^2} = \sqrt{1+1} = \sqrt{2}$$

iv) Calculate the gradient of the line AC .

1

$$m_{AC} = \frac{1-0}{-1+2} = \frac{1}{1} = 1$$

v) Show that the line L_1 makes an angle of 45° with the positive direction of the x -axis.

1

$$m = \tan^{-1} 1 = 45^\circ$$

vi) Find the area of $\triangle ABC$.

2

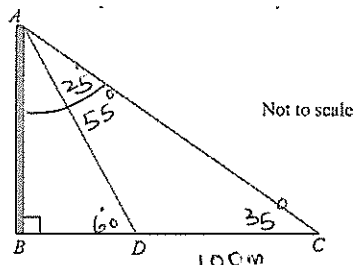
$$\text{Area} = \frac{1}{2} \cdot AB \cdot AC \cdot \sin 45^\circ$$

$$= \frac{1}{2} \cdot (3+2) \cdot \sqrt{2} \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2} \cdot 5 = 2.5 \text{ u}^2$$

Question 11 (13 marks).

- a) A vertical tower AB with points B, C and D in a straight line on the ground is shown below. The distance from CD is 100 metres. The angle of elevation to the top of the tower from point C is 35° and from point D is 60° .



- i) Show that $AD = \frac{100 \sin 35^\circ}{\sin 25^\circ}$ 3

$$\angle CAD = 25^\circ \quad (\text{Ext. } \angle ADB = \angle CAD + 35^\circ)$$

$$\text{Using sine rule in } \triangle ADC, \quad \frac{AD}{\sin 35^\circ} = \frac{100}{\sin 25^\circ}$$

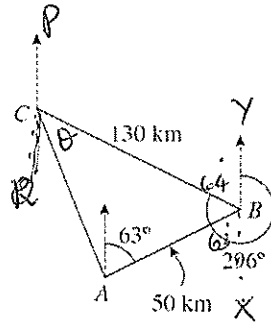
$$\therefore AD = \frac{100 \sin 35^\circ}{\sin 25^\circ}$$

- ii) Calculate the height of the tower. Answer to the nearest metre. 2

$$\sin 60^\circ = \frac{AB}{AD}$$

$$\therefore AB = \frac{100 \sin 35^\circ}{\sin 25^\circ} \times \sin 60^\circ = 117.5367 \dots \approx 118 \text{ m}$$

b) A ship sails 50 km from port A to port B on a bearing of $063^\circ T$, then sails 130 km from port B to port C on a bearing of $296^\circ T$.



i) Show that $\angle ABC = 53^\circ$

2

$$\begin{aligned} \angle ABX &= 63^\circ \quad (\text{alternate } \angle\text{'s on } \parallel \text{ lines}) \\ \angle CBY &= 360 - 296 = 64^\circ \quad (\text{angles at a point}) \\ \therefore \angle ABC &= 180 - (63^\circ + 64^\circ) = 180 - 127 = 53^\circ \end{aligned}$$

ii) Find, correct to the nearest km, the distance of port A from port C.

3

$$\begin{aligned} &\text{Using cosine rule,} \\ AC^2 &= 130^2 + 50^2 - 2 \times 130 \times 50 \cos 53^\circ \\ \therefore AC &= 107.594 \dots = 108 \text{ km} \end{aligned}$$

iii) Use the cosine rule to find $\angle ACB$, and hence find the bearing of port A from port C, correct to the nearest degree.

3

$$\cos \theta = \frac{130^2 + 108^2 - 50^2}{2 \times 130 \times 108} = 0.928$$

$$\theta = 22^\circ$$

$$\begin{aligned} \angle BCP &= 180 - 64 = 116^\circ \\ \therefore \text{bearing of A from port C} &= 116 + 22 = 138^\circ \end{aligned}$$

Question 12.(15 marks).

Prove that $\tan \theta - \tan \theta \sin^2 \theta = \sin \theta \cos \theta$.

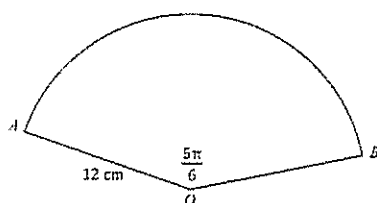
2

LHS $\tan \theta - \tan \theta \sin^2 \theta$

$$= \tan \theta (1 - \sin^2 \theta) = \tan \theta \times \cos^2 \theta$$

$$= \frac{\sin \theta}{\cos \theta} \times \cos^2 \theta = \sin \theta \cos \theta = \text{RHS}$$

b) In the diagram below,



AB is an arc of a circle with centre O . The radius $OA = 12\text{cm}$ and $\angle AOB = \frac{5\pi}{6}$ radians.

i) Find the length of the arc AB .

2

$$l = r\theta$$

$$= 12 \times \frac{5\pi}{6} = 10\pi \text{ cm}$$

ii) Find the area of the sector AOB .

2

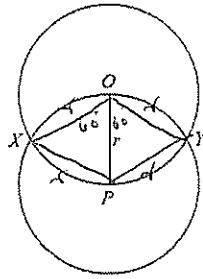
$$\text{Area} = \frac{1}{2} r^2 \sin \theta$$

$$= \frac{1}{2} \times 12^2 \times \sin \frac{5\pi}{6}$$

$$= \frac{1}{2} \times 144 \times \frac{1}{2} = 36 \text{ cm}^2$$

$$= \frac{1}{2} \times 144 \times \frac{5\pi}{6} = 60\pi \text{ cm}^2$$

- b) Two equal circles with centres O and P intersect at X and Y as shown below. The centres of each circle lie on the circumference of the other circle



Find the exact area of the region $XOYP$.

3

Area of segment $XYP = A$

$$A = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} r^2 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$$

$$= \frac{1}{2} r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

Area of the region $XOYP$ is twice the area of segment XYP

$$\therefore A = 2 \times \frac{1}{2} r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) = r^2 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

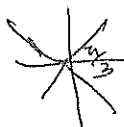
- c) Solve these trigonometric equations in radians, for $0 \leq x \leq 2\pi$.

i) $\sin^2 x = \frac{3}{4}$

ii) $2 \cos^2 x - \cos x - 1 = 0$ (6)

$\sin x = \frac{\sqrt{3}}{2}$ or $\sin x = -\frac{\sqrt{3}}{2}$

related angle $= \frac{\pi}{3}$



$$x = \frac{\pi}{3} \text{ or } \frac{2\pi}{3} \text{ or } \frac{4\pi}{3} \text{ or } \frac{5\pi}{3}$$

Let $u = \cos x$

$$2u^2 - u - 1 = 0$$

$$(2u+1)(u-1) = 0$$

$$u = -\frac{1}{2} \text{ or } 1$$

$$\therefore \cos x = -\frac{1}{2} \text{ or } 1$$

$$\cos x = -\frac{1}{2}, \text{ related } \angle = \frac{2\pi}{3}$$

End of Examination

$$\therefore x = \frac{2\pi}{3} \text{ or } \frac{4\pi}{3}$$

$$\cos x = 1, x = 0 \text{ or } 2\pi$$

$$\therefore x = 0, \frac{2\pi}{3}, \frac{4\pi}{3} \text{ or } 2\pi$$